MathsJam Shout

April 2025
Manchester, UK

Puzzle

The King of the Spiders

The king of the spiders has four servants, and the servants have either 6,7 or 8 legs. Servants with 7 legs always lie, and servants with 6 or 8 legs always tell the truth.

The king asks 'How many legs do you four have in total?', and the four spider servants (who are standing behind a table, so you can't see their legs) answer 25, 26, 27 and 28 respectively.

Who is telling the truth?



Game Farkel

Played using a set of six 6-sided dice, Farkel involves scoring points by rolling combinations of matching dice. You can buy sets which have a special Farkel (⌘) symbol in place of 5 - which has a special significance in the game - but regular dice would also work.

On your turn, you roll all 6 dice, and any which have scored points can be saved, and the remaining dice re-rolled - but if you fail to score, you'll lose all your points for that turn.

Full rules and scoring: **en.wikipedia.org/wiki/Farkle**

See also: **boardgamegeek.com/boardgame/3181/farkle**



Build Post-it note Dodecahedron



Uses 30 square post-it notes.

Video: **youtu.be/vj00hemquss**

See also: **bit.ly/postitdodeca**

Proof

A knot tied in a straight strip of paper, when flattened, makes a pentagon.

Prove that if the sides of the strip are parallel, the pentagon will be regular.

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com** for details.

MathsJam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly MathsJam nights happen in over 30 locations around the world, on the second-to-last Tuesday of each month. To find your nearest MathsJam, visit the website at **www.mathsjam.com**.